

Welcome to the Advanced Machine Learning Course

Course Code: CSA 333

Faculty Introduction

- **Shobhit Johri**
 - AI/ML Faculty at NST- RU
- **Education**
 - B.Tech from IIT Roorkee
- **Previous Experience**
 - Mathematics Research Analyst @ Turing
- **Interests**
 - Machine Learning, Mathematics and Problem Solving





Course Details

- **Course Title:** Advanced Machine Learning
- **Course Code:** CSA 333
- **Credits:** 4
- **Period:** 12 Weeks
- **Evaluation Criteria**
 - Contest: 10% (only 1 contest)
 - Lecture + Lab Assignments: 5%
 - Projects and Viva: 20%
 - Mid Sem: 25%
 - End Sem: 40%
- **Main Course Objective**
 - Build strong ML fundamentals and in depth rigour about ML conceptually, mathematically, and practically.



Major Prerequisites of the Course

- **Mathematics:** Revise key concepts from your previous courses of Probability & Statistics, Linear Algebra and Calculus.
- **Python:** Comfortable with Python programming and commonly used libraries such as NumPy, Pandas, and Matplotlib.
- **Machine Learning:** Familiarity with the fundamentals of ML covered in Introduction to AI & GenAI course.



Major Course Content

- Machine Learning Project Lifecycle
- Linear Regression
- Optimization Techniques
- Regression Analysis
- Bias and Variance
- Feature Engineering: Feature Selection and PCA
- Regularization
- Logistic Regression
- Decision Trees
- Bagging and Boosting
- K Means Clustering
- Advanced Concepts (depending on the course flow)



Class Rules

- 75% attendance compulsory
- Attendance will not be given 10 mins after the start of the lecture and labs. However, the learner may sit inside the lecture for learning.
- Keep your laptops closed inside the class unless instructed otherwise.
- Revise previous lectures and labs before coming to the class.
- Maintain Discipline and Punctuality
- Solve the inclass and post class assignment questions honestly.
- Occupy the front seats first and then go towards back.
- Read, understand and solve your worksheets rigorously and honestly.
- Keep all of your worksheets with yourself.
- Give feedbacks at the end of the lecture on portal.



Thank You

Now Let's move onto Worksheet 0

Setting the Foundation – From Code to Learning

Worksheet for Lecture 0

From Traditional Code to AI, ML & DL - Data Types, Learning Strategies, Core Tasks

Name	Branch / Year	Date

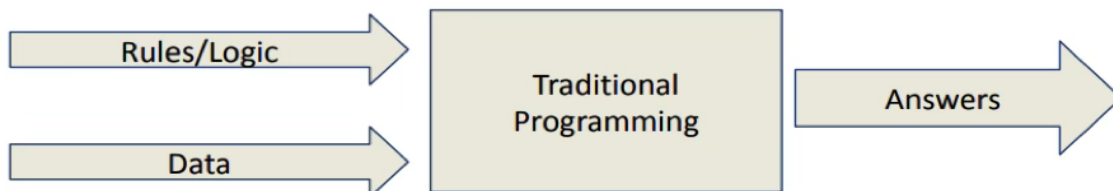
Part 1 The Starting Point: Traditional Programming

HOOK

Imagine you are asked to write a spam filter in Python. You start adding rules: block emails with 'FREE', 'WINNER', 'CRYPTO'. It works - until spammers write 'FR33' or 'W1NNER'. Your program breaks. You patch it. They change it again. What is the fundamental structural problem here?

Definition & Mechanics

Traditional Programming: A paradigm where human engineers manually write explicit logical rules to process input data and produce specific outputs or answers.



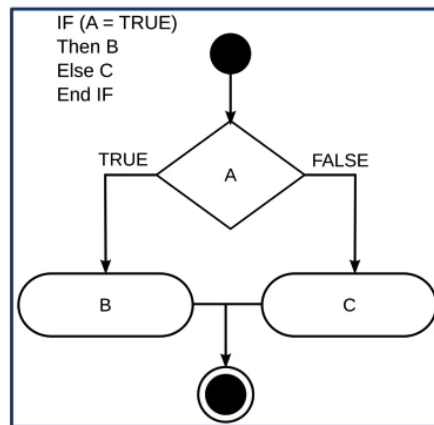
Analogy: Think of traditional programming like writing a rigid cookbook recipe. The developer must completely understand the problem, figure out every rule, every edge case, every logical branch - and explicitly dictate those rules to the computer. The computer is purely an executor with zero autonomy.

The Breakdown: Why This Fails in the Real World

What happens when a spammer realizes their emails are being blocked because of the word "FREE"? They simply change it to "F-R-E-E" or "FR33."

If you rely on traditional programming, your code breaks. To fix it, you have to go back into the source files, manually add another if-else statement, and recompile.

Traditional programs are structurally rigid. They are set in stone. The moment the environment shifts, unusual weather patterns disrupt a logistics system, or users change their slang, the program fails to make appropriate decisions because it cannot adapt dynamically.



REFLECT · Traditional programs are 'structurally rigid - set in stone.' In your own words, what does this mean?

KEY INSIGHT

Traditional programs cannot adapt dynamically. They require manual intervention every time the real world shifts. This rigidity is the core motivation for the entire field of Machine Learning.

TAKEAWAY

Traditional Programming: Data + Rules -> Answers. Hard-coded. Rigid. Breaks when the world changes.

Part 2 The Umbrella: Artificial Intelligence (AI)

HOOK

If rigid, explicit code breaks the moment the real world gets messy - how do humans handle it? How do you tell a spam email from a real one, even if the spammer uses a word you have never seen before? What ability are we trying to give computers?

Building the Intuition

Humans don't run on hardcoded if-else blocks. Human decision-making operates on millions of incredibly flexible, fuzzy, and pattern-based rules that we constantly update. We look at the big picture, notice context, and adapt. If we want computers to diagnose diseases, drive cars, or understand speech, we need them to mimic this broader, generalized adaptability. This aspiration is what we call Artificial Intelligence.

Definition & Core Focus

Artificial Intelligence (AI): The broad, overarching field of computer science dedicated to creating hardware and software systems capable of mimicking human-like intelligence.

The Ultimate Goal: To engineer systems that can independently reason, solve problems, perceive environments, and make autonomous decisions without needing a human to hardcode every single scenario.

Key Areas of Modern AI

AI is an immense field with thousands of specialized sub-domains. Some of the most prominent areas you encounter every day include:

- **Search Algorithms:** Think of Google Maps or Uber routing. It's not just calculating a static line; it's dynamically searching through millions of combinatorial routes to find the optimal path in real-time based on traffic changes.
- **Large Language Models (LLMs):** Systems like the ones you explored last semester that can generate coherent, contextually aware human text.
- **Robotics:** Designing physical agents capable of navigating, manipulating, and interacting safely with the unpredictable physical world.
- **Natural Language Processing (NLP):** The engine behind conversational assistants like Google Assistant or Siri, which take raw audio waves of human speech, parse the grammar, extract semantic meaning, and respond intelligently.
- **Everyday Relatable Examples:** Non-player characters (NPCs) in modern video games that dynamically adapt their combat strategies based on how *you* play the game.
- **Computer Vision (CV):** Training computers to interpret and understand the visual world. By digitizing visual data from analog cameras, systems can utilize deep neural networks to accurately identify, track, and classify objects, faces, or entire scenes.

REFLECT • AI is described as an 'umbrella' field. What does ML fall under, and why is AI broader than just ML?

KEY INSIGHT

AI is the aspiration: machines that mimic human-like adaptability. ML is one of the primary engines used to achieve that aspiration - but AI includes non-learning systems too (e.g., search algorithms).

TAKEAWAY

AI is the overarching field. ML, Search, NLP, CV, Robotics are all sub-domains inside AI.

Part 3 The Engine: Machine Learning (ML)

HOOK

Instead of writing rules, what if we gave the computer 50,000 real emails AND 50,000 spam emails and said: 'Figure out what makes them different'? How does this flip the software engineering paradigm on its head?

The Paradigm Flip

Traditional Programming	Machine Learning
Input: Data + Rules -> Output: Answers	Input: Data + Answers -> Output: Rules

Definition & Mechanics

Machine Learning (ML): A specialized subset of AI focused on developing mathematical models and algorithms that allow computers to autonomously extract patterns and learn insights directly from data, rather than relying on explicit, hardcoded instructions.

The Workflow: The ML model analyzes historical records (training data), identifies implicit statistical relationships, builds a generalized mathematical representation of those relationships, and applies that representation to make predictions or classifications on entirely new, unseen future data.

Classic Examples of ML

Spam Detection: Email servers continuously ingest billions of user actions (marking as spam vs. moving to inbox) to automatically update their internal filtering patterns without software updates.

Stock Price Prediction: Analyzing decades of multi-variable financial tickers, trading volumes, and macroeconomic indicators to forecast future asset pricing trends.

Medical Diagnostics: Evaluating laboratory blood panels, genomic data, and historical patient vitals to predict early onset of chronic illnesses.

KEY INSIGHT

ML models do not need the programmer to know the rules in advance. They generalize from historical labelled examples and apply that generalization to unseen data.

TAKEAWAY

ML: Data + Answers -> Rules. The algorithm discovers patterns; humans supply examples.

Part 4 The Deep Dive: Deep Learning (DL)

HOOK

A basic ML algorithm receives a photo of a cat. The photo is a grid of millions of pixel numbers. If the cat tilts its head, every single pixel value shifts completely. Why does standard ML struggle here - and how does Deep Learning solve it?

Building the Intuition: Structured vs. Unstructured Data

Standard ML is great for...

Structured data: Excel spreadsheets, clean databases
Example features: Square footage, Number of bedrooms

Standard ML struggles with...

Unstructured raw data: images, audio, long essays
Requires human engineers to manually extract features first (e.g., detecting geometric shapes or frequency changes)

The Human Brain Analogy

The human brain handles visual complexity using a massive network of billions of interconnected neurons. When light hits your retina, the first layer of neurons recognizes simple lines and edges. Each subsequent layer builds a more abstract representation. **Deep Learning** does exactly this mathematically.

Definition & Core Strengths

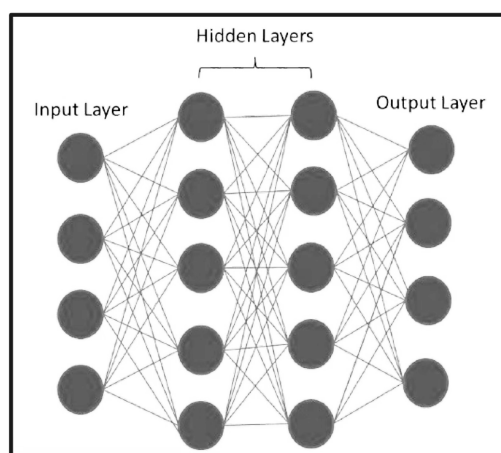
Deep Learning (DL): A highly specialized subfield of Machine Learning that utilizes multi-layered Artificial Neural Networks to automatically discover, extract, and optimize representations directly from raw, unstructured data.

What makes it "Deep"? The term "deep" refers specifically to structural architecture. Instead of a single layer of computation, these models stack dozens or hundreds of processing layers (hidden layers) sequentially.

Key Examples of DL - Complete the 'What DL Does' Column

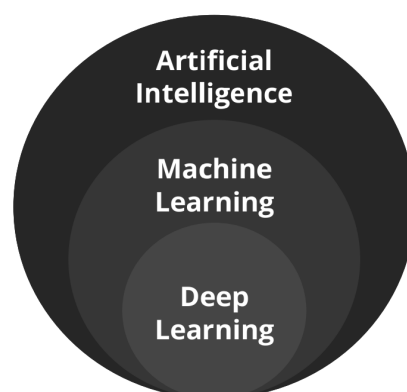
Image Recognition: Autonomous vehicles processing 360-degree video feeds frame-by-frame to instantly identify pedestrians, lane markings, and stop signs.

Voice Assistants: Stripping away background noise from a raw audio waveform to isolate vocal frequencies and transcribe spoken words instantly.



NOTE: You will explore the mathematical foundations of neural propagation, activation functions, and gradient descent in your parallel Deep Learning course.

Visualizing the Ecosystem



KEY INSIGHT

DL solves the feature-engineering bottleneck of standard ML for unstructured data. Depth = many sequential hidden layers, each building more abstract representations than the last.

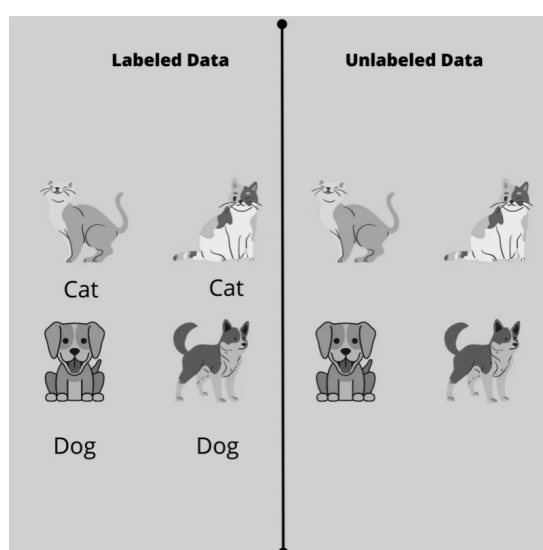
TAKEAWAY

$DL \subset ML \subset AI$. Depth = automatic hierarchical feature learning from raw unstructured data.

Part 5 The Fuel: Labelled vs. Unlabelled Data

HOOK: Suppose I hand you two folders of pet images. In Folder A, every image tells you whether it's a **cat** or a **dog**. In Folder B, you only have the images—no labels, no hints. How would you figure out what's in Folder B?

Building the Intuition



Data comes in two fundamental formats based on whether the 'ground truth answer' is attached to it. If we want a machine to learn, we have to recognize what kind of information we are feeding it - this determines the entire learning paradigm we can use.

The Clear Distinction - Complete Both Definition Boxes

LABELLED DATA: Data that consists of a set of input features coupled directly with the explicit target outcome or correct answer (the ground truth).

UNLABELLED DATA: Raw data that contains only the input features or attributes, with absolutely no accompanying target tags, categories, or historical outcomes.

PRACTICE P3 · Classify Each Dataset

Dataset	Labelled or Unlabelled?
50,000 emails tagged 'spam' or 'not spam'	
200,000 chest X-rays with no radiologist annotations	

REFLECT · Why is labelled data expensive to produce? Use a domain-specific example from the lecture.

KEY INSIGHT

The presence or absence of a ground-truth label determines which entire learning paradigm you use.

Labelling requires human domain experts (radiologists, lawyers, linguists) - which is costly and slow.

TAKEAWAY

Labelled = input + correct answer. Unlabelled = input only. The label determines which paradigm you can use.

Part 6 The Strategies: Four Types of Learning

HOOK

A parent points at a ball and says 'Ball' - teacher-guided learning. The same child left alone on a playground starts exploring, grouping toys by shape. A video game AI gets +10 points for winning, -5 for crashing. How do these three scenarios map to the four ML learning types?

The Four Primary Learning Paradigms

1. Supervised Learning	2. Unsupervised Learning
3. Semi-Supervised Learning	4. Reinforcement Learning

1. Supervised Learning (Teacher-Guided)

Concept: The model learns using labelled data. It provides both the questions (inputs) and the explicit correct answers (labels). The objective is to learn a general mapping function that can accurately predict the label for new, unseen inputs.

Example: Predicting housing market prices. Features: square footage, location. Label: exact sale price.

2. Unsupervised Learning (Self-Organized Discovery)

Concept: The model is fed completely unlabelled data. No teacher, no scoring key. The algorithm acts like a data detective - uncovering hidden structures, natural groupings, and underlying patterns on its own based on statistical similarities.

Example: Market segmentation. An algorithm discovers distinct consumer archetypes from anonymous customer purchasing histories.

3. Semi-Supervised Learning (The Practical Hybrid)

Concept: In the real world, labelling data is incredibly expensive and time-consuming because it requires human experts. Semi-supervised learning addresses this by combining a very small pool of high-quality **labelled data** with an incredibly massive pool of cheap, raw **unlabelled data**. The model uses the small labelled dataset to anchor its basic understanding, then propagates that knowledge across the vast unlabelled dataset.

CASE STUDY - GOOGLE PHOTOS

When you open your photo app, it automatically clusters faces together (unsupervised step). Then it asks: "Is this person Alex?" - one single label from you. It instantly labels thousands of other photos containing that same face (supervised propagation).

4. Reinforcement Learning (Trial & Error Paradigm)

Concept: Does not rely on static datasets. An autonomous agent interacts dynamically with an unknown environment. The agent takes an action, experiences consequences, and receives feedback: rewards (positive outcomes) or penalties (mistakes). Over millions of iterations it learns to optimize its policy to maximize cumulative rewards.

Why 'unknown environment' matters: The agent has no map, no prior knowledge, and no dataset to draw from. It must explore entirely through trial and error - generating its own training signal by interacting with a world it has never seen. This is what makes RL fundamentally different from the other three paradigms.

CASE STUDY - AUTONOMOUS DRONE

Training a drone to fly through windy industrial warehouses. The drone gets penalized when it crashes and rewarded when it maintains stability and reaches its destination. Over millions of simulated flights, it learns the optimal rotor adjustment policy.

KEY INSIGHT
Supervised = labelled data, learns mapping.
Unsupervised = unlabelled, discovers structure.
Semi-supervised = real-world compromise: labels are expensive.
Reinforcement = no dataset; agent explores an unknown environment and generates its own feedback.
TAKEAWAY: Four paradigms - one key question: Do you have labels? Does your agent need to explore an unknown environment to generate its own feedback?

Part 7 The Objectives: Core AI Tasks
Regression · Classification · Clustering

HOOK: Uber needs to set a surge multiplier (1.4×, 2.1×...). A hospital needs to flag a patient as 'high risk' or 'low risk'. Netflix needs to group viewers into micro-genre clusters with no predefined categories. Three problems. Three different task types. How do you tell them apart instantly?

Now that we understand the strategic paradigms of learning, let’s catalog the concrete engineering tasks we execute in the industry. Every complex machine learning system ultimately distills down to one or more of these core operations/tasks:

Task	Output Type	Supervised or Unsupervised?
Regression	Continuous numerical value (scalar)	Supervised
Classification	Discrete predefined category / label	Supervised
Clustering	Unlabelled natural group	Unsupervised

Task A: Regression - Predicting Continuous Values

What is it? Regression is a supervised learning task where the target output variable is completely continuous and numerical. Instead of picking from a set of discrete categories, the model outputs a precise scalar value along an infinite mathematical continuum.

When do we need it? We need regression whenever our prediction target answers questions like *"How much?"*, *"How many?"*, or *"What value?"*.

Real-World Applications

- **House Price Prediction:** Calculating the exact valuation of a home based on square footage, zip code, and historical trends.
- **Temperature Forecasting:** Predicting tomorrow’s peak afternoon temperature in Celsius.
- **Car Mileage Estimation:** Determining expected miles per gallon (MPG) based on engine displacement and chassis weight.

CASE STUDY - UBER SURGE PRICING
Uber utilises sophisticated real-time regression architectures to analyze localized demand metrics, active driver distributions, weather constraints, and time variables to dynamically output a continuous fare multiplier (e.g., 1.4×, 2.1×) for specific geo-fenced zones.

Common Regression Algorithms

1. Linear Regression	2. Support Vector Regression (SVR)
3. Decision Tree Regression	4. Random Forest Regression

Task B: Classification - Predicting Categories

What is it? The target output is a discrete, predefined category or class label. The objective is to draw clear decision boundaries in the data space to separate different classes cleanly.

When do we need it? Whenever the target answers: "Is this A, B, or C?", "Is this safe or unsafe?", "Which species does this belong to?"

Real-World Applications - Complete the Table

Application	Input Data	Output Label / Category
Spam vs. Not Spam	Email payload content	A binary choice evaluating incoming email payloads.
Disease Diagnostics	Clinical vitals, blood biomarkers	Scanning clinical vitals to classify a patient as positive or negative for a condition.
Computer Vision Labelling	Isolated image pixel matrix	Cat / Dog / Fox (label the specific class)

CASE STUDY - MEDICAL SCREENING

Hospitals deploy specialized classification models to cross-reference patient symptoms, blood biomarkers, and respiratory vitals to immediately categorize emergency room admissions into probability risk brackets for infectious diseases.

Common Classification Algorithms

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree Classification
- Support Vector Machines (SVM)
- Naive Bayes Classifier
- Random Forest Classification

Task C: Clustering - Grouping Similar Data

What is it? An unsupervised learning task. The algorithm analyzes natural statistical distributions, dense regions, and geometric proximity of data points to group them into distinct, coherent clusters. Points inside a cluster are highly similar to each other, and highly distinct from points in other clusters.

When do we need it? When we don't know the categories in advance and want to discover hidden structures, segmentations, or natural divisions inside raw, unstructured customer or operational data.

Real-World Applications

- **Customer Behavior Segmentation:** Identifying hidden sub-groups of users based on purchasing patterns to design tailored marketing campaigns.
- **Document Topic Modeling:** Scanning thousands of academic papers to automatically organize them into thematic subject groups without human reading.

- **Image Segmentation:** Partitioning an image into distinct object regions by clustering pixels with similar color channels and spatial proximity. Unlike classification (which labels the whole image), segmentation labels every pixel - producing a map of what each region of the image contains.

CASE STUDY - NETFLIX MICRO-GENRES

Netflix utilises complex clustering algorithms to evaluate user viewing behaviors, watch histories, and interactions. Instead of using broad categories like 'Sci-Fi', these algorithms uncover hundreds of highly specific, localized user taste-clusters to drive personalized homepage recommendation carousels.

Common Clustering Algorithms

- **K-Means Clustering**
- **K-Medoids Clustering**
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**
- **Hierarchical Clustering (Agglomerative)**
- **Gaussian Mixture Models (GMM)**

Summary Wrap-Up

Today, we built a coherent narrative of our field. We looked at how traditional programming breaks under real-world complexity, necessitating the adaptive umbrella of Artificial Intelligence. We zoomed into Machine Learning to understand how systems learn rules directly from data, and explored how Deep Learning mimics human neural structures to handle complex, unstructured data. Finally, we categorized our toolkits into structural data flavors (Labelled vs. Unlabelled), learning strategies (Supervised, Unsupervised, Semi-Supervised, Reinforcement), and concrete engineering tasks (Regression, Classification, Clustering).